

University for the Creative Arts

School of Games and Creative Technology

BSc (Hons) Computer Science

FGCT6021 MOBILE APPLICATION DEVELOPMENT

LAB 5 – TOPIC 5 PYTHON & JSON BACKEND DATA

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Week: 5

Lab 5 – Week 5 Journal Summary (What You Have Done)

1. Setup Python Environment in VS Code

- You downloaded a portable version of Python (python-3.13.7-embed-amd64.zip).
- Extracted and used the portable Python folder in VS Code without requiring admin installation.
- Installed the Python extension in VS Code and configured it to use the portable interpreter.
- Created and ran a test script (example.py) successfully.

2. Created Backend JSON Storage

- You created a folder named backend with:
 - reflections.json → initialized with an empty array: [].
 - save_entry.py → A Python script to write reflections to JSON.

3. Python Integration

- Wrote save_entry.py to:
 - Accept journal input from the user via the terminal.
 - Read the existing reflections.json content.
 - Append new entries with timestamp.
 - Save it back to reflections.json.

4. PWA Integration

- Your Learning Journal PWA:

- Uses JavaScript (fetch API) to read from reflections.json and display reflections.
- Displays them dynamically inside the journal page.
- Uses DOM manipulation to add each entry to the page.
- Includes basic error handling and interactivity.

5. Additional Feature

- You added an extra feature: a reflection counter and an export button that allows users to download the reflections.json file, improving user experience and backup options.

6. Weekly Journal Answered (Reflected in Code and HTML)

- You wrote and displayed answers to:
 - Difference between JSON vs browser storage.
 - How Python was used to update JSON.
 - What the PWA displays locally vs on GitHub.
 - What extra feature was added and why.